CMOS HALF-ADDER: VLSI DESIGN, SIMULATION AND LAYOUT IN MICROWIND

Internship Report

Submitted by

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for the award of the degree of

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in

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Velagapudi Ramakrishna Siddhartha Engineering College
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CERTIFICATE

This is to certify that the Internship report entitled '***CMOS HALF-ADDER: VLSI DESIGN, SIMULATION AND LAYOUT IN MICROWIND***' was prepared and presented by **Sugrivu Durga Bhavani (228W1A0457)** of B. Tech, 8th semester, in Electronics and Communication Engineering, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Communication Engineering at Velagapudi Ramakrishna Siddhartha Engineering College, affiliated with Jawaharlal Nehru Technological University, Kakinada, during the academic year 2025-2026.

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DECLARATION

I, **Sugrivu Durga Bhavani (228W1A0457)** of B. Tech 4th year in Electronics and Communication Engineering at Velagapudi Ramakrishna Siddhartha Engineering College, Kanuru, Vijayawada, here by declare that this internship has been carried out at '**SkillDzire**' for a period of 12 Weeks and the report has been submitted in partial fulfilment for the requirement of B. Tech curriculum.

Place: ………………………………

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Abstract

This report presents the work carried out during a four-month internship at SkillDzire, focusing on the principles of VLSI design and chip architecture within the ASIC design flow. The internship provided a comprehensive understanding of digital circuit design, modeling, simulation, and physical layout implementation used in integrated circuit development. As part of the practical component, a Half Adder circuit was designed and implemented to demonstrate the complete VLSI design process. The circuit was modeled and verified using Verilog HDL, followed by transistor-level implementation and layout design using the Microwind VLSI CAD tool. Emphasis was placed on ensuring functional correctness through pre-layout and post-layout verification, including Design Rule Check (DRC) and Layout Versus Schematic (LVS) validation. The project also explored important design aspects such as area optimization, power consumption, and propagation delay, highlighting the relationship between logical design and physical implementation. This hands-on experience bridged theoretical concepts with practical VLSI workflows enhancing technical proficiency in digital circuit

About The Company

SkillDzire is an emerging ed-tech platform based in Hyderabad, India, focused on providing industry-oriented skill development programs, internships, and professional training. Established in 2020, the organization aims to bridge the gap between academic learning and industry requirements by equipping students with practical and job-ready skills. The platform offers a wide range of courses and internship programs across multiple domains such as Artificial Intelligence, Data Science, VLSI, and Web Development. These programs are designed and delivered by industry experts, ensuring hands-on learning through real-time projects. SkillDzire follows a learner-centric approach by combining theoretical knowledge with practical exposure. It provides certification, project-based learning, and placement assistance to enhance students' career opportunities. Additionally, the organization offers structured internship programs that enable students to work on real-world projects, thereby improving their technical skills and understanding of industry practices.

CHAPTER 1

INTRODUCTION

1.1 Background

The rapid advancement of semiconductor technology has led to the development of highly efficient and compact integrated circuits, making VLSI (Very Large Scale Integration) a fundamental aspect of modern electronics. VLSI design involves integrating millions of transistors onto a single chip, enabling the implementation of complex digital systems used in applications such as communication devices, computing systems, and embedded technologies. Among the basic building blocks of digital systems, arithmetic circuits play a crucial role. The Half Adder is one of the simplest combinational circuits used to perform binary addition of two input bits, generating Sum and Carry outputs. Despite its simplicity, it serves as a foundational element for designing more complex circuits such as full adders and arithmetic logic units (ALUs).

The VLSI design process includes several stages such as design entry, functional verification, synthesis, and physical layout design. Tools like Verilog HDL are used for modeling and simulation, while Microwind enables the implementation of transistor-level layouts and analysis of circuit performance.

This project focuses on the design and implementation of a Half Adder using VLSI design techniques, providing hands-on exposure to both digital logic design and physical layout development. It also emphasizes the importance of verification, optimization, and adherence to design rules for achieving efficient circuit performance.

1.2 Objectives

The main objective of this project is to design and implement a Half Adder using VLSI design principles and analyze its performance through simulation and layout verification. The work aims to provide practical exposure to digital circuit design and the complete VLSI design flow.

- To design a Half Adder using basic digital logic gates.
- To model and verify the circuit using Verilog HDL.

- To implement the transistor-level design using CMOS technology.
- To develop the physical layout using Microwind VLSI CAD tool.
- To perform DRC and LVS verification for design accuracy.
- To analyze performance parameters such as area, power, and delay.

1.2 Scope of Work

This project focuses on the design and implementation of a Half Adder using VLSI design principles, covering both front-end and back-end stages of the design flow. It includes functional modeling and verification of the circuit using Verilog HDL, followed by transistor-level implementation using CMOS technology. The physical layout of the circuit is developed using the Microwind VLSI CAD tool, ensuring compliance with fabrication rules through Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification. The project also involves pre-layout and post-layout simulations to validate circuit performance and analyze key parameters such as area, power consumption, and propagation delay. Overall, the scope emphasizes gaining practical knowledge of digital circuit design, layout development, and verification in VLSI systems.

1.3 Organization of Report

This report is organized into four chapters, each dealing with a specific stage of the internship work and project implementation.

Chapter 1: This chapter introduces the background of VLSI design and chip architecture, along with the importance of arithmetic circuits in digital systems. It presents the objectives, scope of work, and overview of the CMOS Half Adder project.

Chapter 2: This chapter explains the methodology followed during the internship. It describes the VLSI design flow, Half Adder design using XOR and AND gates, Verilog HDL modeling, CMOS transistor-level implementation, and physical layout development using Microwind.

Chapter 3: This chapter presents the results and discussion of the project. It includes layout implementation, DRC and LVS verification, input-output analysis, timing analysis, and post-layout simulation results. The performance of the CMOS Half Adder is evaluated in terms of area, power, and delay.

Chapter 4: This chapter concludes the report by summarizing the work carried out during the internship, the outcomes achieved, and the knowledge gained in VLSI design, layout development, and verification. It also discusses the future scope of the project.

1.4 Summary

Chapter 1 introduces the importance of VLSI (Very Large Scale Integration) in modern electronics, where millions of transistors are integrated onto a single chip to build compact and efficient digital systems. It explains that arithmetic circuits are essential in digital design and highlights the Half Adder as a basic combinational circuit that adds two binary inputs to generate Sum and Carry outputs. The chapter also notes that the Half Adder serves as the foundation for more advanced circuits such as full adders and arithmetic logic units (ALUs).

The chapter further describes the VLSI design flow, including design entry, functional verification, synthesis, and physical layout design. Verilog HDL is used for modeling and simulation, while Microwind is used for transistor-level layout and performance analysis. The objective of the project is to design and implement a CMOS Half Adder, verify its correctness, create its physical layout, and analyze parameters such as area, power consumption, and propagation delay.

CHAPTER 2

METHODOLOGY

2.1 Approach and Implementation

The methodology adopted during the internship was structured to follow the principles of VLSI design and chip architecture, covering both front-end and back-end design stages. The work focused on designing, implementing, and verifying a Half Adder circuit using a systematic VLSI design flow. The objective was to gain practical understanding of digital circuit design, physical layout, and performance evaluation using industry-relevant tools such as Verilog HDL and the Microwind VLSI CAD tool. The following approach was followed:

1. **Understanding VLSI Design Flow:** The work began with studying the complete VLSI design flow, including specification, RTL design, functional verification, synthesis, and physical design. This provided a clear understanding of how digital circuits are transformed from high-level descriptions into physical layouts.
2. **Design of Half Adder Circuit:** A Half Adder was designed using fundamental digital logic concepts. The circuit performs binary addition of two input bits and generates Sum and Carry outputs using XOR and AND gates.
3. **Verilog HDL Modeling and Simulation:** The designed Half Adder was described using **Verilog HDL** at the behavioral and structural levels. Functional simulation was performed to verify correctness for all possible input combinations and ensure accurate output generation.
4. **Transistor-Level Implementation:** The verified logic design was translated into a transistor-level representation using CMOS logic. Basic gates such as XOR and AND were implemented using PMOS and NMOS transistors.
5. **Layout Design using Microwind:** The transistor-level circuit was implemented as a physical layout using the Microwind tool. This involved designing individual gate layouts, interconnections, and optimizing the placement for area efficiency while following fabrication constraints.

6. **Design Rule Check (DRC):** The layout was verified using DRC to ensure that all design rules related to spacing, width, and layers were satisfied, making the design suitable for fabrication.
7. **Layout Versus Schematic (LVS) Verification:** LVS checks were performed to confirm that the layout accurately represents the intended schematic design, ensuring functional equivalence.
8. **Post-Layout Simulation and Performance Analysis:** Post-layout simulations were carried out to analyze the impact of parasitic elements on circuit behavior. Performance parameters such as propagation delay, power consumption, and area were evaluated to assess design efficiency.

This structured methodology enabled a clear understanding of the complete VLSI design cycle, from digital logic design to physical implementation. The hands-on experience with modeling, layout design, and verification strengthened the ability to develop efficient and reliable integrated circuits within modern chip architecture frameworks.

2.2 Summary

Chapter 2 describes the methodology followed to design and implement the CMOS Half Adder using the VLSI design flow. The work began with understanding the complete VLSI process, including specification, RTL design, functional verification, synthesis, and physical design. A Half Adder circuit was then designed using XOR and AND logic to generate the Sum and Carry outputs.

The circuit was modeled and simulated in Verilog HDL to verify its correctness for all possible input combinations. After functional verification, the design was converted into a transistor-level CMOS implementation using PMOS and NMOS transistors. The physical layout of the circuit was then created in Microwind, where transistor placement and interconnections were optimized for compact area and proper functionality.

The layout was verified using Design Rule Check (DRC) to ensure compliance with fabrication rules and Layout Versus Schematic (LVS) verification to confirm that the layout matched the intended circuit design. Finally, post-layout simulation was performed to evaluate the effect of parasitic components and to analyze important performance parameters such as propagation delay, power consumption, and area.

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Result Analysis

The internship at SkillDzire provided practical exposure to VLSI design, focusing on digital circuit modeling and layout implementation. A Half Adder was designed and verified using Verilog HDL, ensuring correct Sum and Carry outputs. The circuit was implemented at the transistor level and its layout was developed using the Microwind tool.

Verification through Design Rule Check (DRC) and Layout Versus Schematic (LVS) confirmed design accuracy. Performance analysis showed that parameters such as delay, power, and area are influenced by layout and interconnections. Overall, the work demonstrated the effective transition from logic design to physical implementation in VLSI systems.

3.1.1 Layout Implementation and Verification

The Half Adder was realized using complementary NMOS and PMOS transistors, implementing the XOR function for the Sum output and the AND function for the Carry output. The layout adhered to standard design rules and was optimized to minimize area while ensuring correct interconnections. Post-layout simulations confirmed that the circuit produced the correct Sum and Carry outputs for all possible input combinations, validating functional correctness.

The Half Adder was implemented in Microwind at the transistor level, with careful consideration of layout optimization and connectivity. The layout diagram represents the physical placement of NMOS and PMOS transistors, interconnections through metal layers, and proper insertion of contacts and vias to ensure correct functionality. The circuit diagram depicts the logic implementation, with an XOR gate producing the Sum output and an AND gate producing the Carry output. Post-layout simulations confirmed that the designed circuit correctly generates Sum and Carry for all input combinations, while the layout adheres to CMOS design rules, maintains compact area utilization, and minimizes parasitic effects.

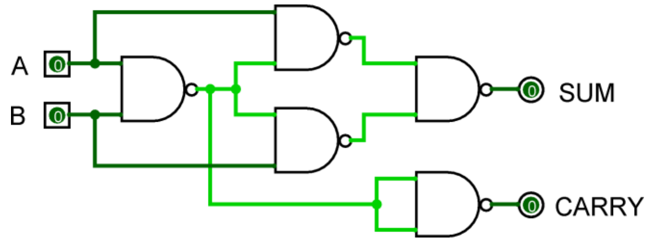

Figure 3.1: Half Adder using NAND Gates

Figure 3.1 shows the circuit diagram of the Half Adder implemented using only NAND gates. Inputs A and B are first applied to a NAND gate, and the intermediate outputs are combined through additional NAND gates to generate the Sum output. The Sum output behaves as an XOR function, while the Carry output is obtained through a NAND-based AND function.

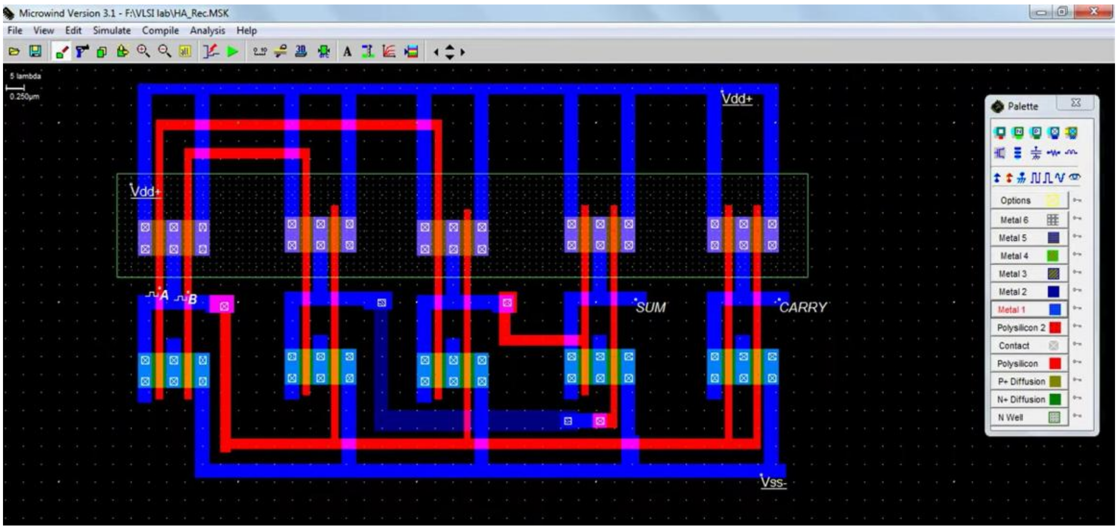

Figure 3.2: Layout Design of half Adder in MIcrowind

NAND gates, while Figure 3.2 presents the corresponding layout design in Microwind, illustrating the physical placement of NMOS and PMOS transistors, interconnections, and contacts used to realize the circuit at the transistor level.

3.1.2 Input-Output Verification

Table 3.1 confirms the input-output verification confirms that the designed Half Adder produces correct Sum and Carry outputs for all possible combinations of inputs A and B. The Sum output follows the XOR operation, while the Carry output follows the AND operation. Simulation results in Microwind matched the theoretical truth table exactly.

Table 3.1: Input-Output Verification of Half Adder

Input A	Input B	Sum Output ($A \oplus B$)	Carry Output ($A \cdot B$)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

3.1.3 Timing Analysis

The designed layout was evaluated for area efficiency and signal propagation delay. Optimization of transistor sizing and placement effectively reduced parasitic capacitances, contributing to faster switching and lower delay. The compact design ensures scalability for integration into more complex combinational logic circuits.

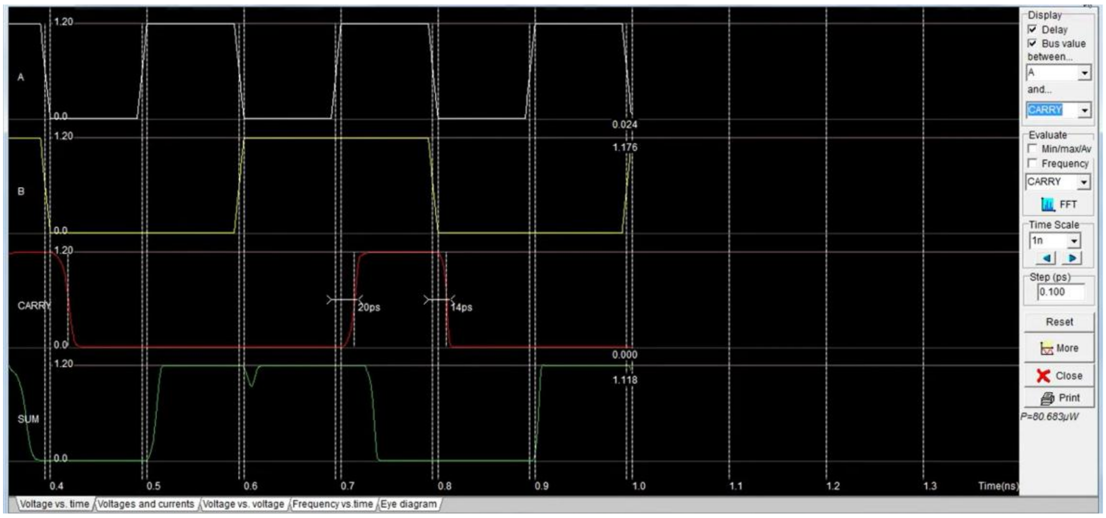

Figure 3.3: Output Waveforms of Layout Design

Figure 3.3 shows the simulated output waveform of the Half Adder in Microwind, illustrating the correct Sum ($A \oplus B$) and Carry ($A \cdot B$) outputs for all input combinations.

The waveform confirms that the circuit operates as expected, with the Sum output high when the inputs are different and the Carry output high only when both inputs are high.

3.2 Significance of the Results

The results obtained in this project demonstrate the successful conversion of a simple digital logic circuit into a complete CMOS-based VLSI implementation. The correct operation of the Half Adder verifies the effectiveness of the design methodology and confirms that the circuit can be reliably used as a building block for more complex arithmetic units such as Full Adders and ALUs. The DRC and LVS verification results are significant because they ensure that the designed layout is both fabrication-ready and functionally equivalent to the intended schematic. The compact layout and reduced propagation delay indicate that proper transistor sizing and placement can improve performance and reduce area usage in integrated circuits. Furthermore, the post-layout analysis highlights the practical importance of considering parasitic effects during phys-ical design. The project provides valuable insight into how layout decisions affect power, delay, and overall efficiency in VLSI systems, thereby strengthening understanding of modern semiconductor design practices.

3.3 Summary

Chapter 3 presents the results obtained from the design and implementation of the CMOS Half Adder. The circuit was successfully modeled, simulated, and implemented using Microwind. Functional verification confirmed that the Half Adder produced the correct Sum and Carry outputs for all input combinations. The layout design followed CMOS rules and was validated using Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification. The chapter also discusses the physical implementation of the Half Adder using PMOS and NMOS transistors. The generated layout was compact and efficient, with proper interconnections and minimum parasitic effects. Post-layout simulation waveforms confirmed that the circuit behaved correctly after layout implementation. In addition, the design was evaluated in terms of area, propagation delay, and power consumption, showing that optimized transistor placement and sizing improved overall circuit performance.

CHAPTER 4

CONCLUSION AND FUTURE WORK

4.1 CONCLUSION

The internship provided significant practical exposure to VLSI design and chip architecture, enabling a clear understanding of the complete design flow from behavioral modeling to physical layout implementation. The successful design and implementation of the Half Adder using Verilog HDL and Microwind demonstrated the effective translation of digital logic into CMOS-based transistor-level circuits. The project emphasized the importance of systematic design methodology, including functional verification, layout design, and validation through Design Rule Check (DRC) and Layout Versus Schematic (LVS). It also highlighted how physical design aspects influence key performance parameters such as propagation delay, power consumption, and area. Overall, the internship bridged the gap between theoretical concepts and practical application, enhancing technical proficiency in digital circuit design, layout development, and verification. This experience provides a strong foundation for future work in VLSI system design and semiconductor technology.

4.2 FUTURE WORK

The future work of this project can focus on extending the Half Adder design to more complex arithmetic circuits such as Full Adders, Ripple Carry Adders, and Arithmetic Logic Units (ALUs). The design can also be optimized further to reduce power consumption, propagation delay, and chip area using advanced CMOS techniques. In addition, the layout may be implemented using smaller technology nodes to improve performance and integration density. Future work can also include FPGA implementation and comparison with other VLSI CAD tools for better analysis and verification.

References

[1] Bhattacharya, Debaprasad, *Digital VLSI Design and Test*, McGraw-Hill, 2008.

[2] Chandrasekaran, K., and R. S. S. Prasad, ”Design and Layout of CMOS Logic Gates and Combinational Circuits,” *International Journal of Electronics and Communication Engineering*, 2018.

[3] Kang, Sung-Mo, and Yusuf Leblebici, *CMOS Digital Integrated Circuits: Analysis and Design*, 3rd Edition, McGraw-Hill, 2003.

[4] Rabaey, Jan M., ”Digital Integrated Circuits: Design, Fabrication, and Test,” *IEEE Transactions on VLSI Systems*, vol. 9, no. 3, pp. 299–311, 2001.

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CERTIFICATE OF INTERNSHIP

This is to Certify that Mr./Ms

Sugrivu Durga Bhavani

Student of Electronics And Communication Engineering

From the college Velagapudi Ramakrishna Siddhartha Engineering College

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VLSI Design & Chip Architecture

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